
LECTURE NOTE

Lecture 3: Networking Fundamentals

AI510 - Cybersecurity and Innovation

Simon Holm
Fall - 2026



Contents

1	Computer networks	3
2	Network topology	3
3	Links and Packets	4
4	Key functions	5
4.1	Forwarding and routing	5
4.2	Packet switching	5
4.2.1	Store and forward	5
4.2.2	Queueing	6
4.3	Alternative to packet switching: Circuit switching	6
5	The Internet	7
6	Network Protocol	7
6.1	The layers	8
7	The four sources of Packet Delay	9
8	Throughput	10

1 Computer networks

Definition: Network Definition by NIST

A system implemented with a collection of interconnected components. Such components may include routers, hubs, cabling, telecommunications controllers, key distribution centres, and technical control devices.

by NIST SP 800-172

2 Network topology

- Connected computers
 - Hosts: end systems
 - Network edges
- Packet switches
 - Routers and switches forward data
- Communication links
 - Any medium: copper, radio, etc.
 - Transmission rate: bandwidth

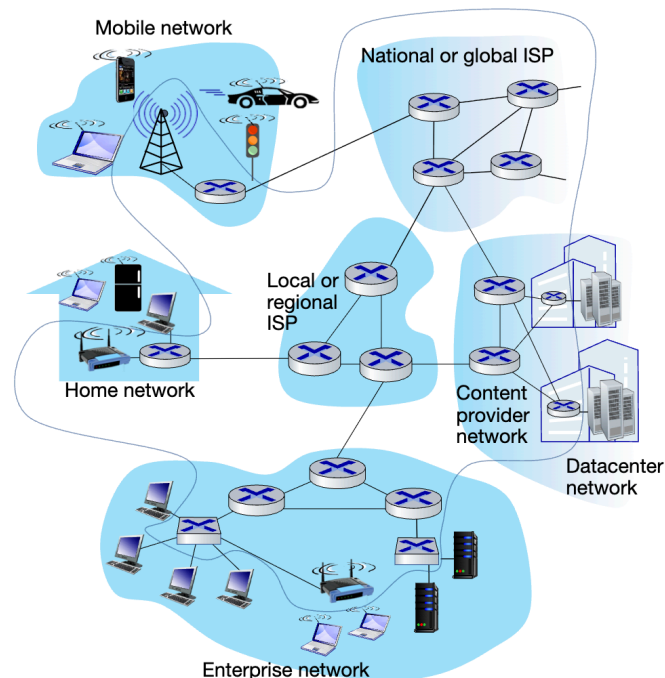


Figure 1: Network example

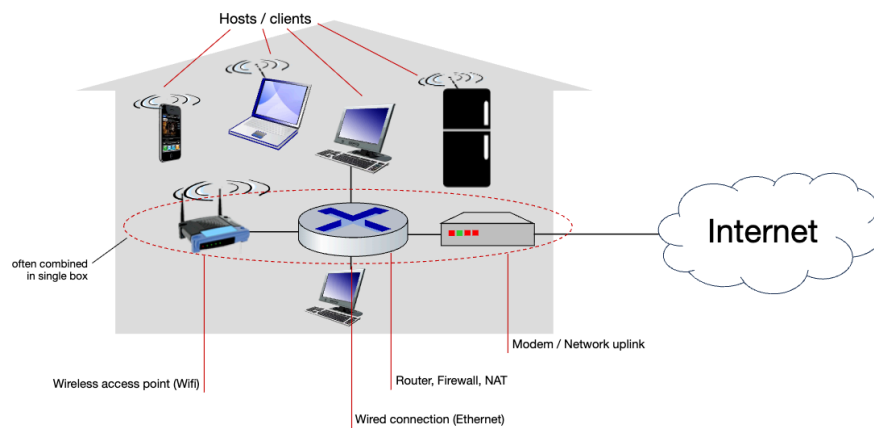


Figure 2: The topology of a typical home network

3 Links and Packets

Hosts break messages up in 'packets' with length L in bits. These are then transmitted at rate R . Then the 'delay' is defined as $\text{delay} = \frac{L(\text{bits})}{R(\text{bits/sec})}$

Example: Shared network access: Cable

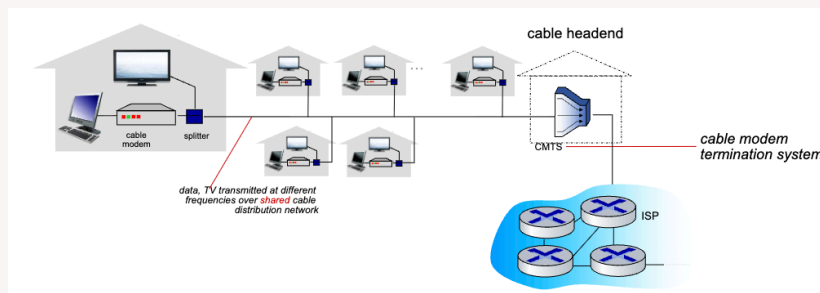


Figure 3: Uses Hybrid Fiber Coax

Example: Dedicated network access: DSL

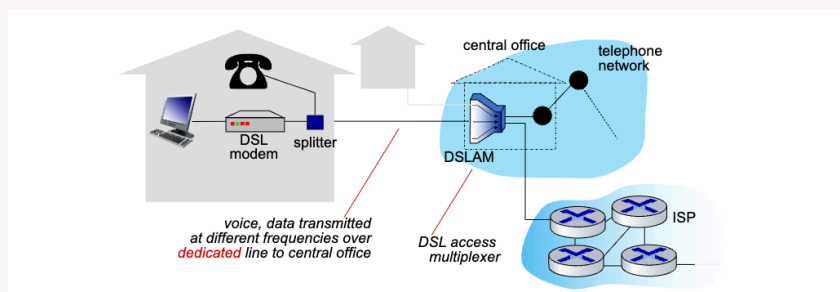


Figure 4: Uses copper wires used in traditional landlines, signal is split

Example: Enterprise networks

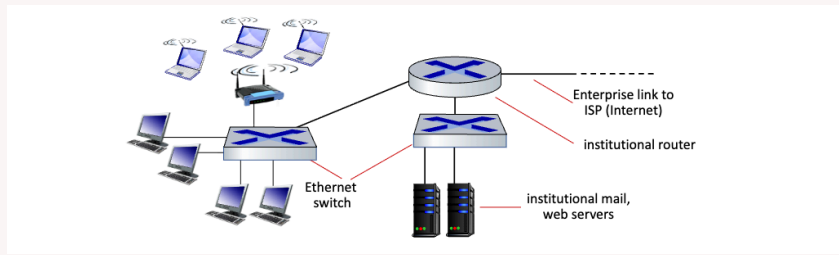


Figure 5: Accommodate requirements of companies, universities, etc

Example: Data centre networks

- High bandwidth links are essential
- Mostly ethernet-based connections, up to currently 400G

4 Key functions

4.1 Forwarding and routing

- Forwarding is a local action, moving packets from the routers input-link to the appropriate output link (*switching*)
- Routing is a global action which determines source-destination paths taken by packets
 - Different routing algorithms

4.2 Packet switching

4.2.1 Store and forward

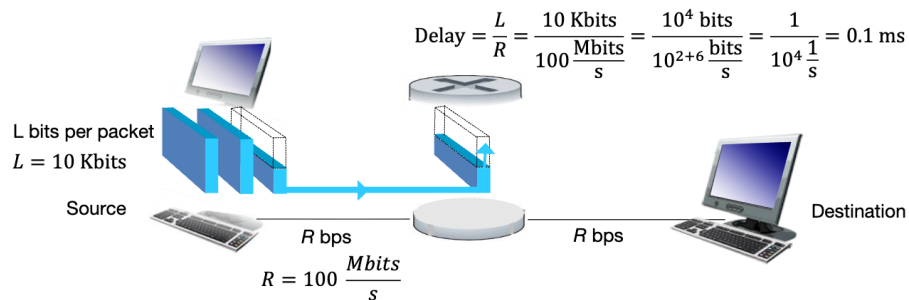


Figure 6: Takes $\frac{L}{R}$ seconds to transmit L -bit packet into link at R bps

The entire packet must arrive at router before it can be transmitted on next link

4.2.2 Queueing

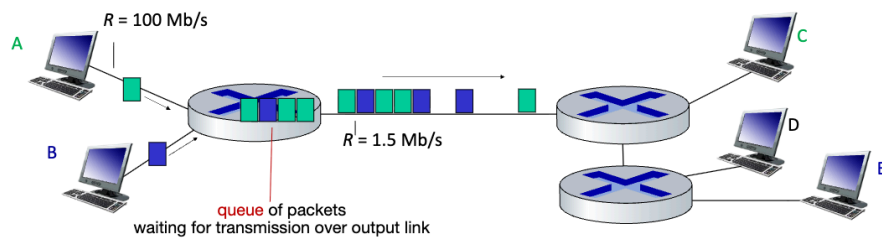


Figure 7: Occurs if work arrives faster than it can be served

If the system cannot store enough we will lose packets (data loss)

4.3 Alternative to packet switching: Circuit switching

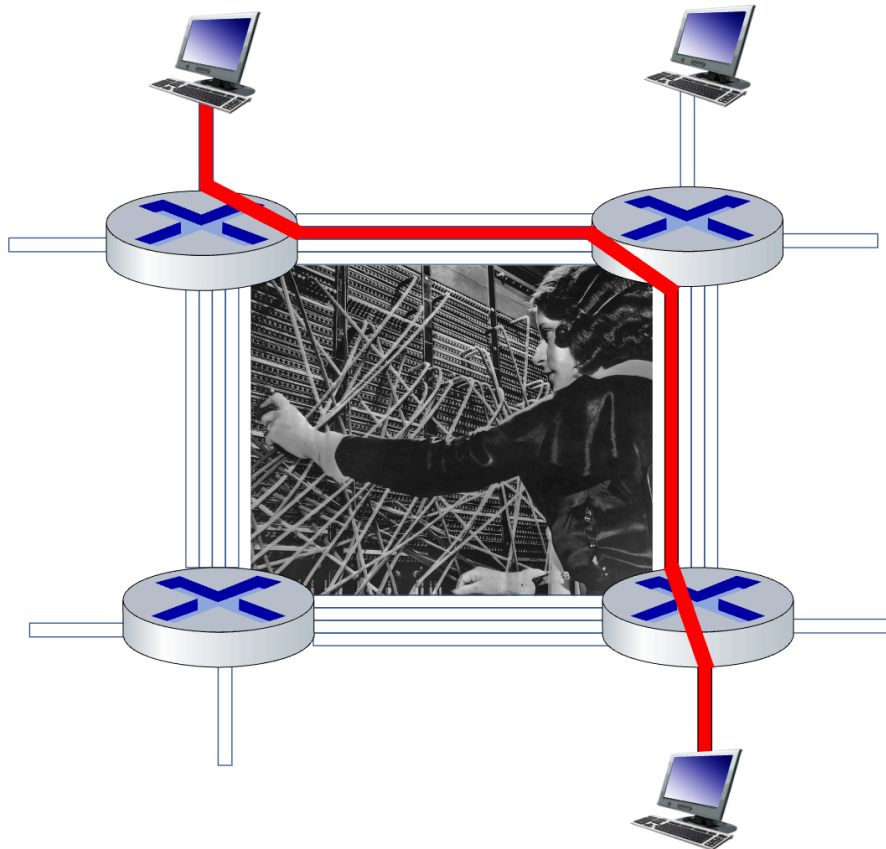


Figure 8: Requires exclusive allocation of end-to-end resource. Segments remain idle if bandwidth not fully used

- Frequency Division Multiplexing (FDM)
 - Each connection allocated its own band, can transmit at max. rate of that band
- Time Division Multiplexing (TDM)
 - Each call allocated periodic slots, can transmit at maximum rate of wider band

Example: Packet Switching vs Circuit Switching

- Packet switching is more flexible and great for “bursty” data
 - Resource can be shared when not needed
 - Simpler, no call setup
- Circuit Switching is better to deliver reliable data transfer
 - No packet loss due to congestion
- Is it possible to provide circuit-like behaviour with packet switching?
 - Kinda, QoS techniques exist

5 The Internet

On the internet, different hosts connect via ISP's. Given the number networks to interconnect. The best way to do it

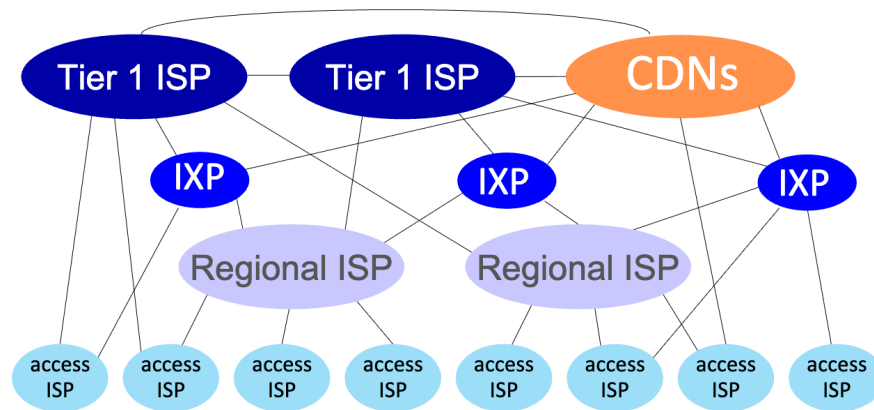


Figure 9: A network of networks

- Tier 1 ISP which can connect everyone
- CDNs (Content delivery network) for companies like ‘facebook’
- IXP (DIX in denmark) are Internet exchange point for shortcuts
- Regional ISPs
- Access ISPs

6 Network Protocol

Definition: Network Protocol

A protocol defines the format and the order of messages sent and received among network entities, as well as the actions taken on message transmission and/or receipt.

by J.F. Kurose & K.W. Ross, Computer Networking

They specify rules for how to format messages and which actions should be taken once messages are received or other events take place

Example: Intuitive understanding of network protocols

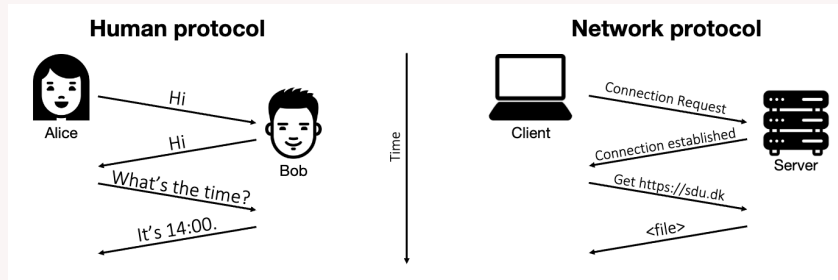


Figure 10: We do this as humans as well.

Protocols are structured as layers. This eases maintenance and updating of the system

Definition: Air Travel

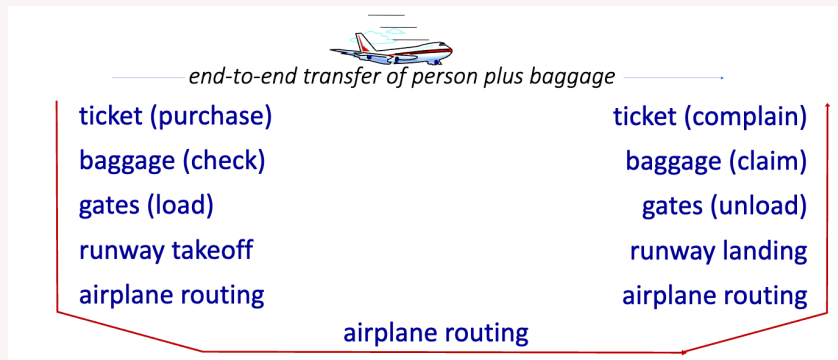


Figure 11: Much like air travel. We encounter much of the same protocol layers going in, as going out

6.1 The layers

- **Application Layer:** Supporting Network Applications
 - HTTP, IMAP, SMTP, DNS
- **Transport Layer:** Process-process data transfer
 - TCP, UDP
- **Network Layer:** Routing of datagrams from source to destination
 - IP, routing protocols
- **Link layer:** Data transfer between neighboring network elements
 - Ethernet, 802.11 (Wifi)
- **Physical Layer:** Bits "on the wire"

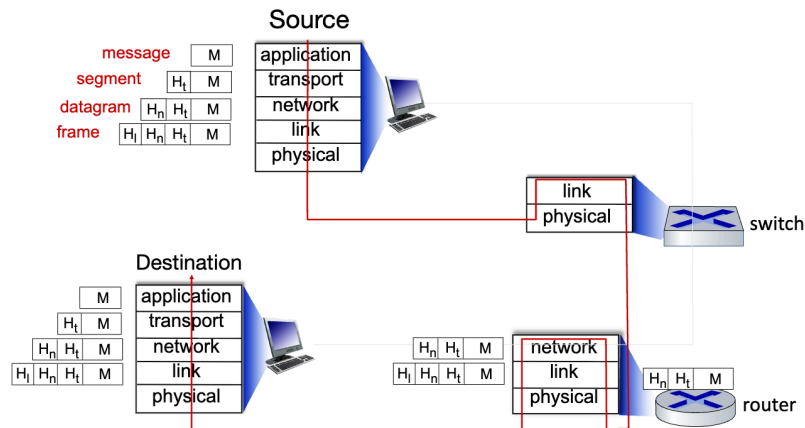


Figure 12: Encapsulation - an end-to-end example

Data is ‘encapsulated’ through each layer. At each source layer data is wrapped in a ‘head’ which can be unwrapped again by the destination.

The ISO/OSI reference model

Some books/internet sources might reference the “ISO/OSI” reference model. However we have since its creation removed the presentation and session layer.

WILL NOT BE IN THE EXAM

7 The four sources of Packet Delay

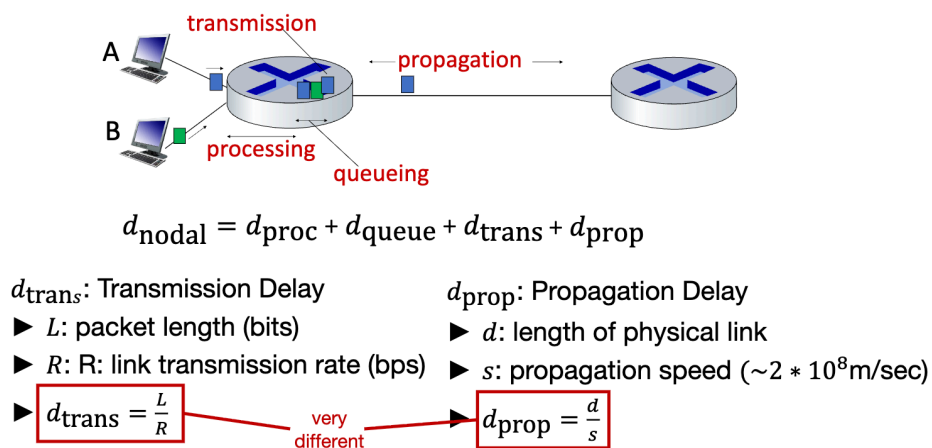


Figure 13:

Example: A car analogy

Given 10 cars (aka 10-bit packet) where the toll takes 12 sec and propagation of 1 hour:

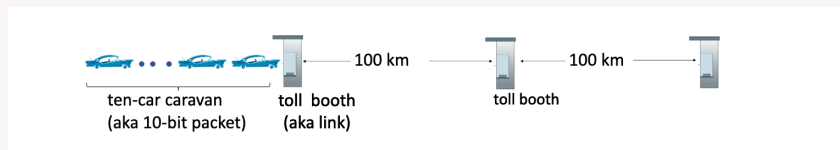


Figure 14: full travel will take 62 minutes

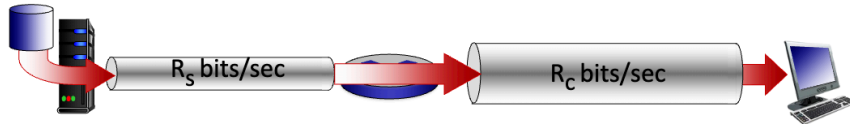
See this using

```
1 traceroute google.com
```

terminal

8 Throughput

$R_s < R_c$: What is average end-to-end throughput?



$R_s > R_c$: What is average end-end throughput?

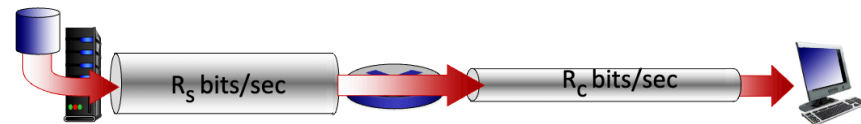


Figure 15: Bottleneck link: The smallest link constraints end-end throughput